

Network Topology and Switching

Course Code: **COE 3201**

Course Title: **Data Communication**

**Dept. of Computer Engineering
Faculty of Engineering**

Lecture No:	11	Week No:	13

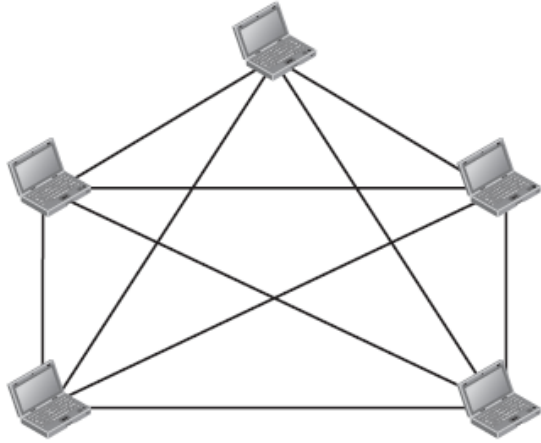
Lecture Outline

1. Network topologies
2. Switching Mechanism for Data Transfer
3. Circuit Switching
4. Message Switching
5. Packet Switching
6. Virtual Circuit Switching
7. Datagram Switching

Network topologies

Figure 1.4 *A fully connected mesh topology (five devices)*

$n = 5$
10 links.



Advantages:

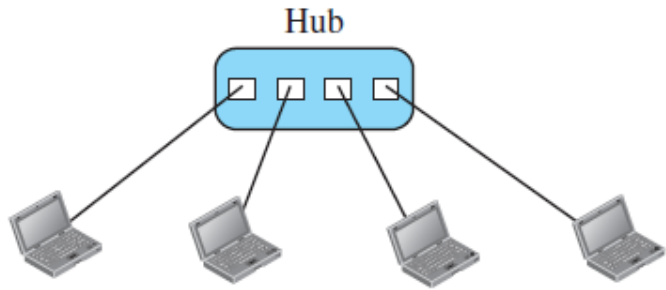
- Communication is very fast between the nodes.
- Mesh Topology is robust.
- The fault is diagnosed easily. Data is reliable because data is transferred among the devices through dedicated channels or links.
- Provides security and privacy.

Disadvantages:

- The cost of cables is high as bulk wiring is required, hence suitable for less number of devices.
- Installation and configuration are difficult.
- The cost of maintenance is high.

Network topologies

Figure 1.5 *A star topology connecting four stations*



Disadvantages:

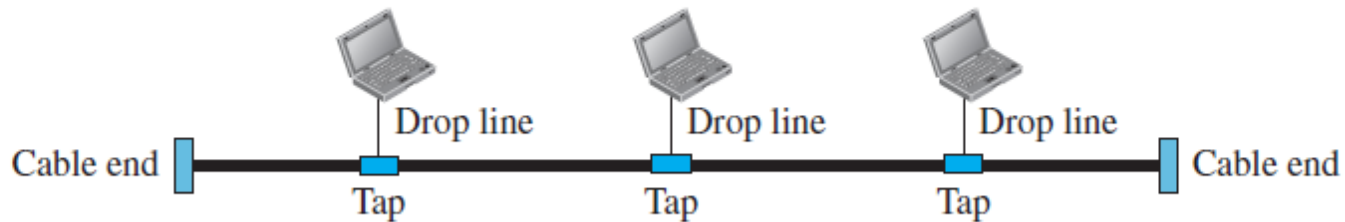
- If the concentrator (hub) on which the whole topology relies fails, the whole system will crash down.
- The cost of installation is high.
- Performance is based on the single concentrator i.e. hub.

Advantages:

- If N devices are connected to each other in a star topology, then the number of cables required to connect them is N . So, it is easy to set up.
- Each device requires only 1 port i.e. to connect to the hub, therefore the total number of ports required is N .
- It is Robust. If one link fails only that link will affect and not other than that.
- Easy to fault identification and fault isolation.
- Star topology is cost-effective as it uses inexpensive coaxial cable.

Network topologies

Figure 1.6 *A bus topology connecting three stations*



Advantages:

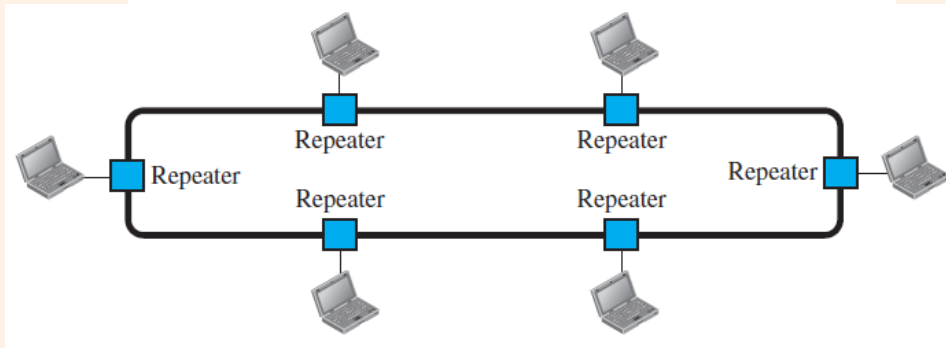
- **Cost-effective:** Requires less cable than other topologies.
- **Easy to install:** Simple layout with minimal cabling.
- **Best for small networks:** Works well in environments with limited devices.

Disadvantages:

- **Limited scalability:** Performance degrades as more devices are added.
- **Single point of failure:** Failure of the main bus cable brings down the entire network.
- **Difficult troubleshooting:** Hard to detect the fault source.
- **Low security:** All nodes receive all transmissions.

Network topologies

Figure 1.7 *A ring topology connecting six stations*



Advantages:

- **Efficient for handling high traffic:** Data flows in one direction (or both in dual ring), reducing collisions.
- **Easy fault identification:** Each node checks the signal, helping detect where a fault occurs.
- **Equal access:** All nodes have equal opportunity to transmit.

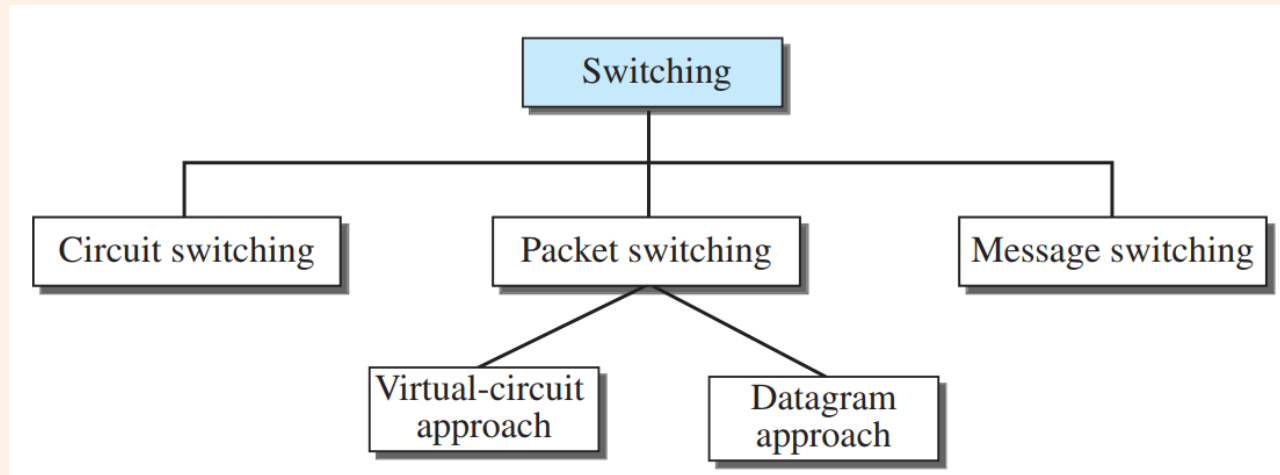
Disadvantages:

- **Single point of failure:** A failure in any one cable or device disrupts the whole network (unless a dual ring is used).
- **Slower with many devices:** Each message must pass through intermediate nodes.
- **Difficult to reconfigure or troubleshoot:** Any change requires halting the network.

Switching Mechanism for Data Transfer

Taxonomy of Switched Network

Traditionally, three methods of switching have been discussed: **circuit switching**, **packet switching**, and **message switching**. The first two are commonly used today. The third has been phased out in general communications but still has networking applications. Packet switching can further be divided into two subcategories—virtual circuit approach and datagram approach—as shown in Figure. Here, we discuss only circuit switching and packet switching; message switching is more conceptual than practical.



Circuit Switching

- ❑ Dedicated communication path between two stations
- ❑ Must have switching capacity and channel capacity to establish connection
- ❑ Must have intelligence to work out routing
- ❑ Inefficient
 - ✓ Channel capacity dedicated for duration of connection
 - ✓ If no data, capacity wasted
- ❑ Set up (connection) takes time
- ❑ Developed for voice traffic (phone)
- ❑ Examples
 - ✓ Telephone networks
 - ✓ ISDN (Integrated Services Digital Networks)

Circuit Switching

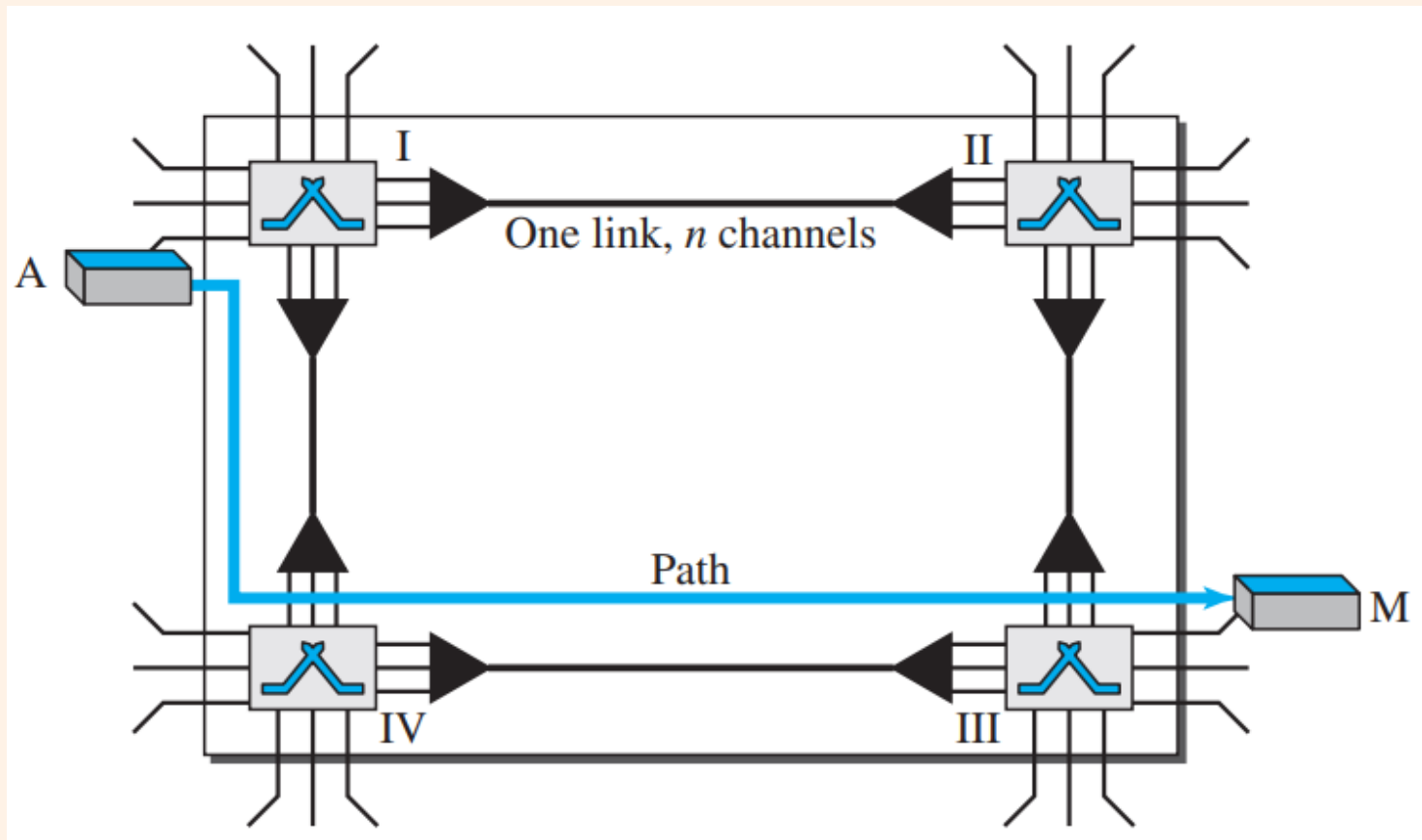


Figure: A trivial circuit-switched network

Circuit Switching

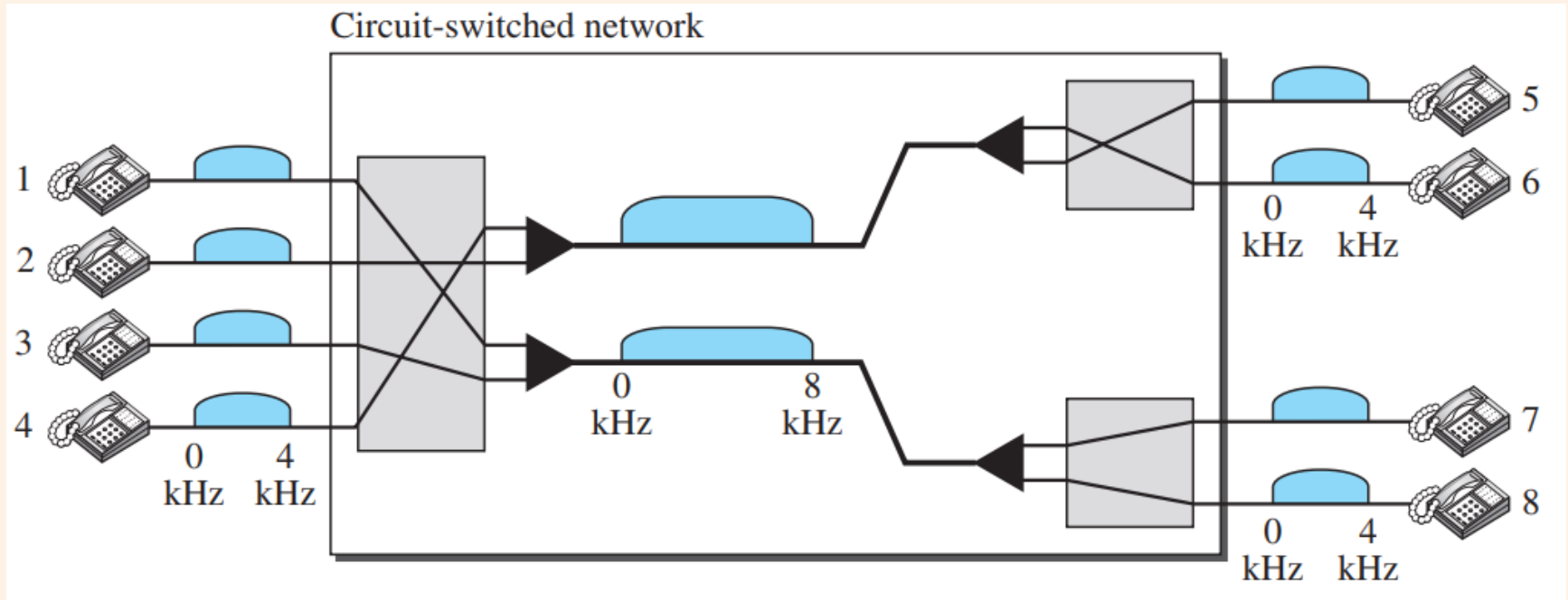


Figure: Circuit-switched network used in Example 8.1

Example 8.1

As a trivial example, let us use a circuit-switched network to connect eight telephones in a small area. Communication is through 4-kHz voice channels. We assume that each link uses FDM to connect a maximum of two voice channels. The bandwidth of each link is then 8 kHz. Figure 8.4 shows the situation. Telephone 1 is connected to telephone 7; 2 to 5; 3 to 8; and 4 to 6. Of course the situation may change when new connections are made. The switch controls the connections.

Circuit Switching

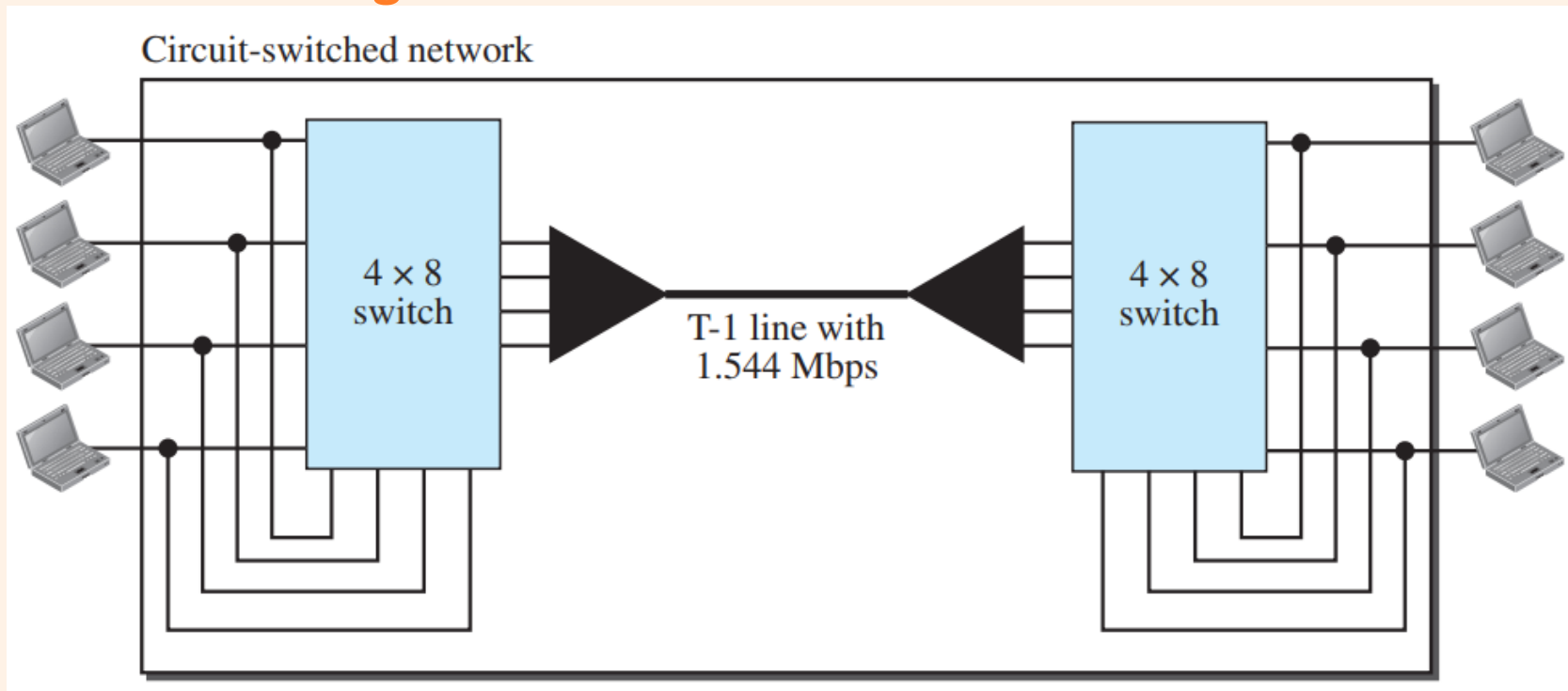
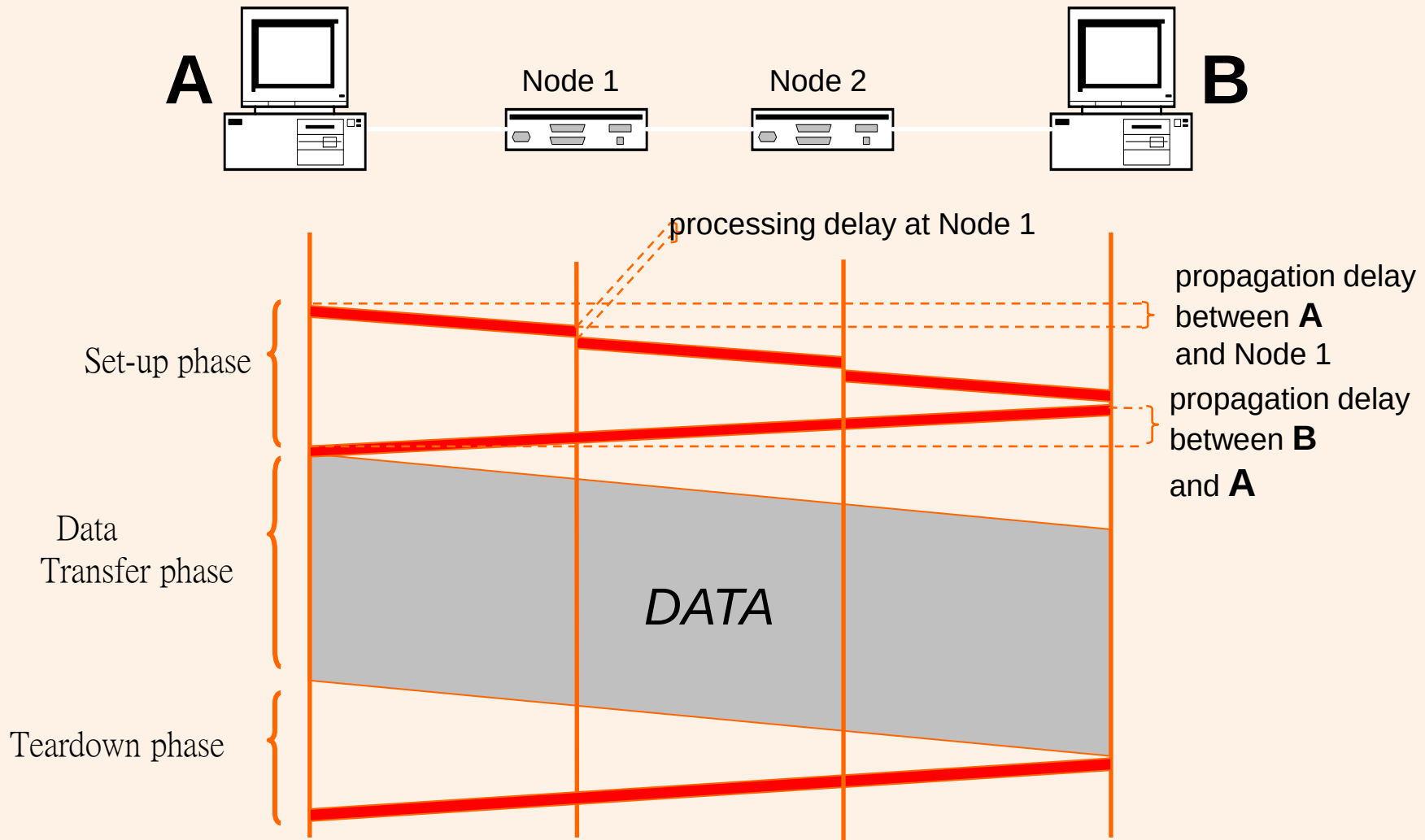


Figure: Circuit-switched network used in Example 8.2

Example 8.2

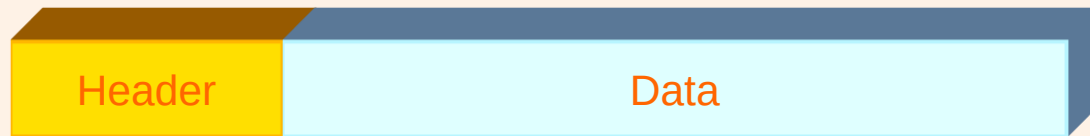
As another example, consider a circuit-switched network that connects computers in two remote offices of a private company. The offices are connected using a T-1 line leased from a communication service provider. There are two 4×8 (4 inputs and 8 outputs) switches in this network. For each switch, four output ports are folded into the input ports to allow communication between computers in the same office. Four other output ports allow communication between the two

Delay in Circuit Switching



Message Switching

- ❑ No dedicated path needs to be established between end-nodes.
- ❑ Source and destination node do not interact in real time. There is no need to determine the status of the destination node before sending the message.
- ❑ Each message is an independent entity and carries address information of the destination. There is no upper limit on the size of the message.



- ❑ The messages are stored at each node before being forwarded to the next node in the route.
- ❑ Message switching accept all traffic but offers longer delivery time than circuit switching. Circuit switching blocks/rejects access traffic.

Packet Switching

- ❑ Messages are broken into small segments of bit-sequences and they are called packets. As packets are restricted to a specific size, they can be routed more rapidly.
- ❑ Packets have the following structure:

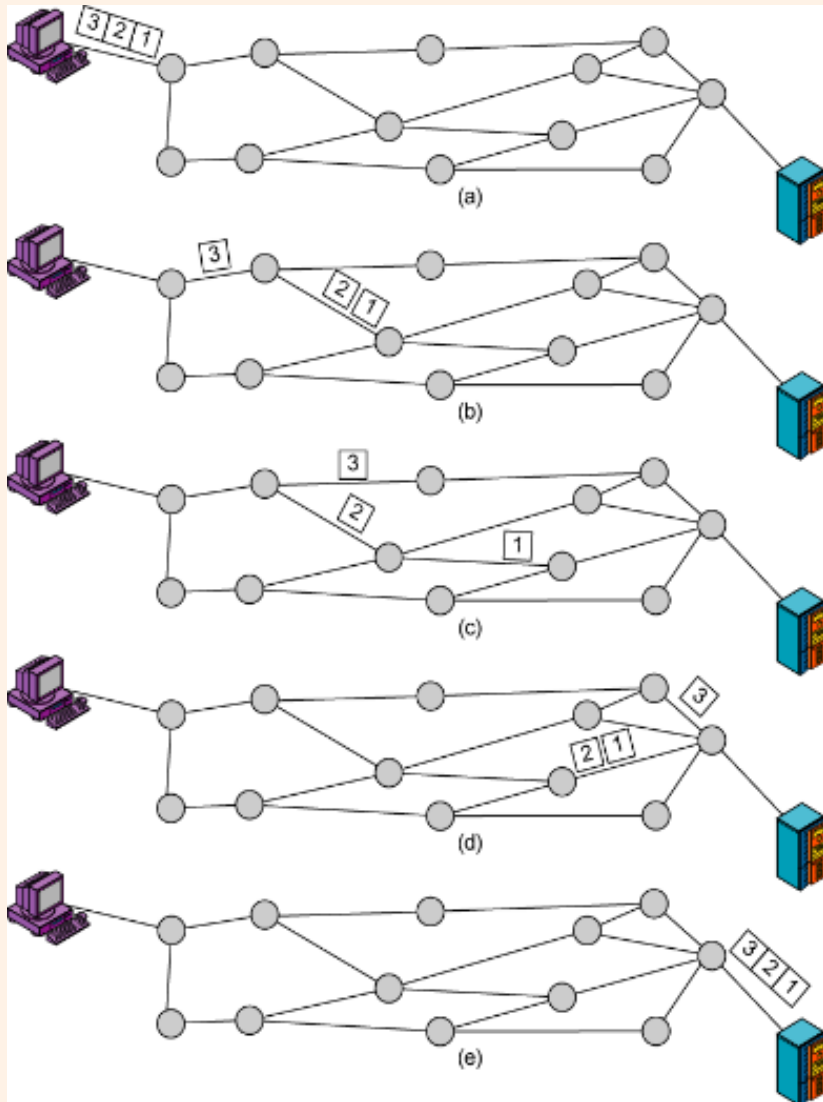


- ✓ Header carries control information (e.g., destination id, source id, message id, packet id, control info)
- ❑ Each packet is passed through the network from node to node along some path (**Routing**)
- ❑ At each node the entire packet is received, stored briefly, and then forwarded to the next node (**Store-and-Forward Networks**)
- ❑ Typically, no storage is required at nodes/switches for packets.

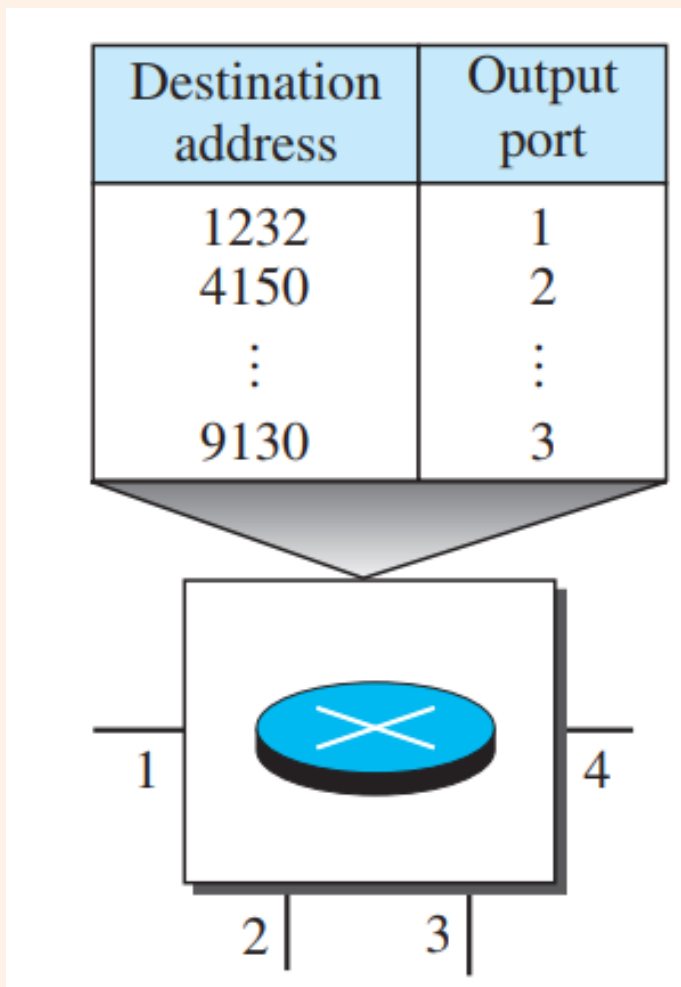
Packet Switching Advantages

- ❑ Packetization allows short messages to get through a transmission link without waiting behind long messages.
- ❑ Line efficiency
 - Single node to node link can be shared by many packets over time
 - Packets queued and transmitted as fast as possible
- ❑ Nodes can accept packet and keep it in the packet queue as long as they have enough space in buffer memory of queue.
 - But it may increase queueing delay
- ❑ Packet priority can be used in the packet queue to introduce network QoS

Datagram packet switching



- Each packet is independently switched
 - Each packet header contains destination address
- No resources are pre-allocated (reserved) in advance
- Routes may change during session



- ❑ A switch in a datagram network uses a routing table that is based on the destination address.
- ❑ The destination address in the header of a packet in a datagram network remains the same during the entire journey of the packet.

Figure: Routing table in a datagram network

Delay in Datagram packet switching

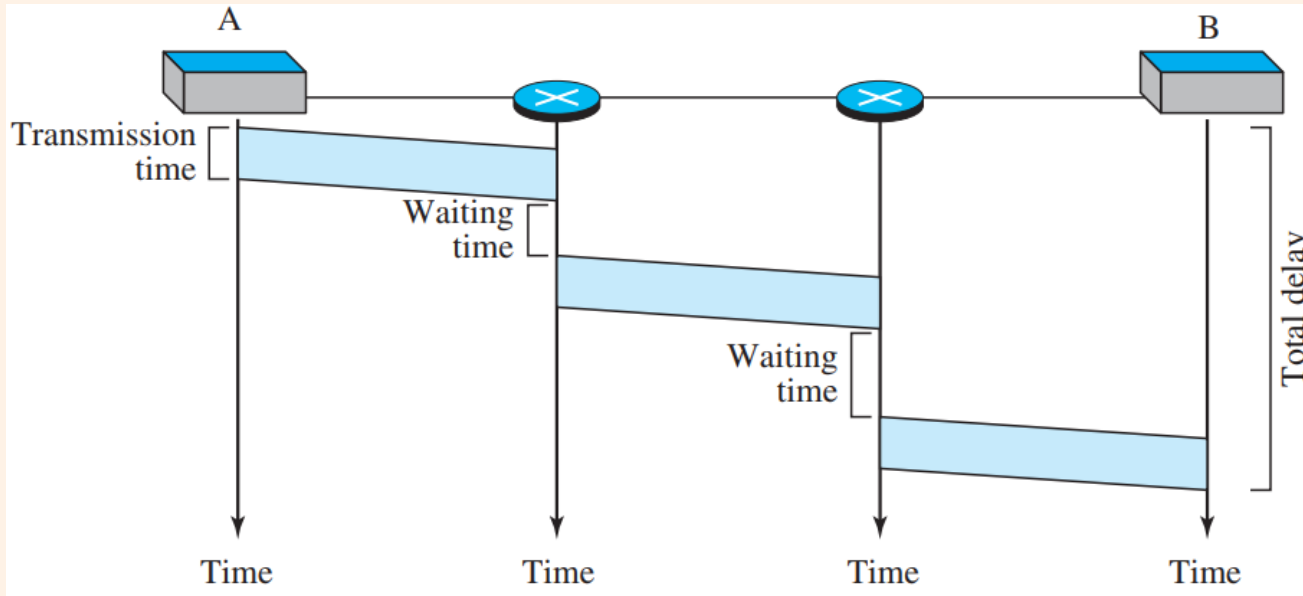


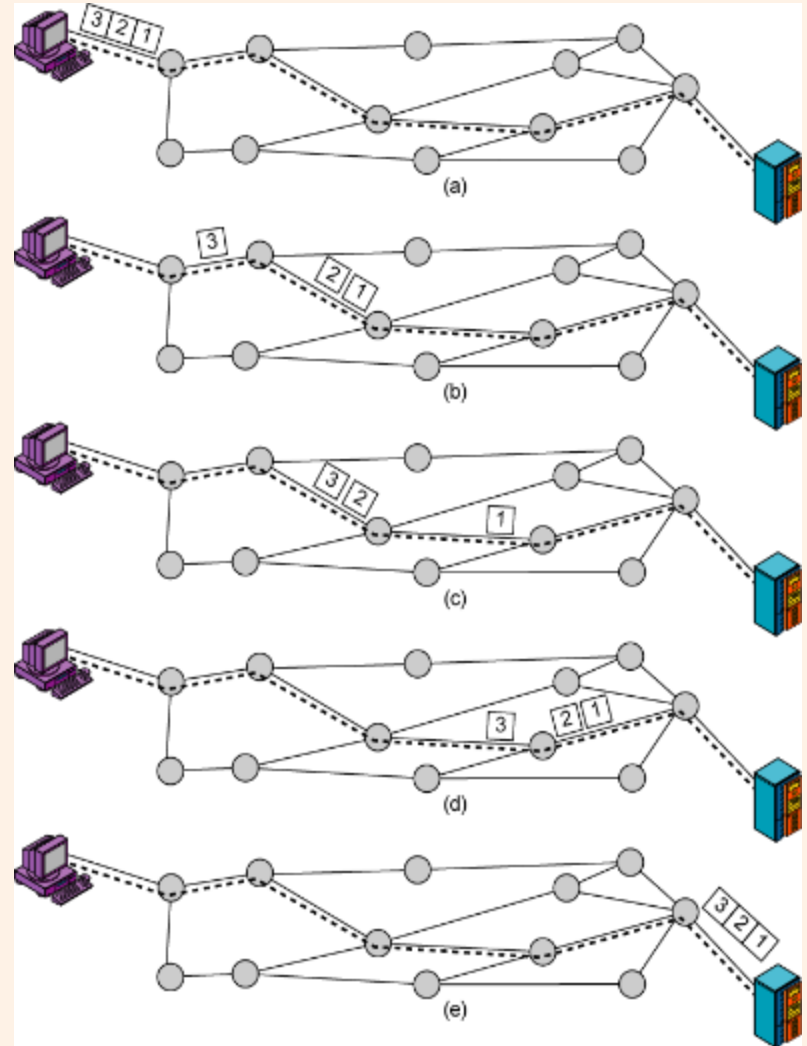
Figure: Delay in a datagram network

The packet travels through two switches. There are three transmission times ($3T$), three propagation delays (slopes 3τ of the lines), and two waiting times ($w_1 + w_2$). We ignore the processing time in each switch. The total delay is

$$\text{Total delay} = 3T + 3\tau + w_1 + w_2$$

Virtual-Circuit Packet Switching

- ❑ A virtual-circuit network is a cross between a circuit-switched network and a datagram network.
- ❑ Preplanned route established before any packets sent
- ❑ Resources can be allocated: predefined/on-demand
- ❑ Call request and call accept packets establish connection (handshake)
- ❑ Communication with virtual circuits takes place in three phases
 - VC establishment
 - data transfer
 - VC disconnect



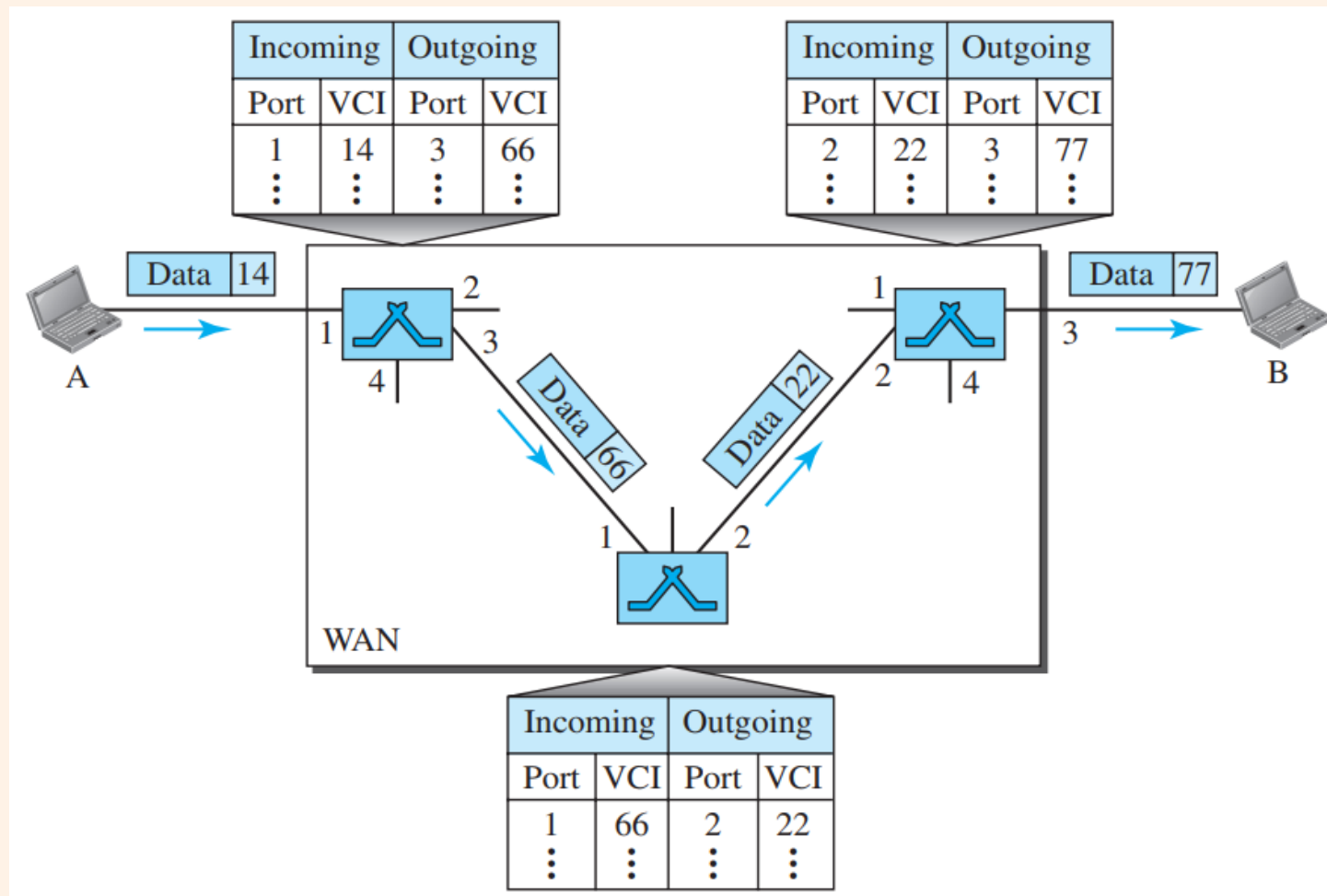
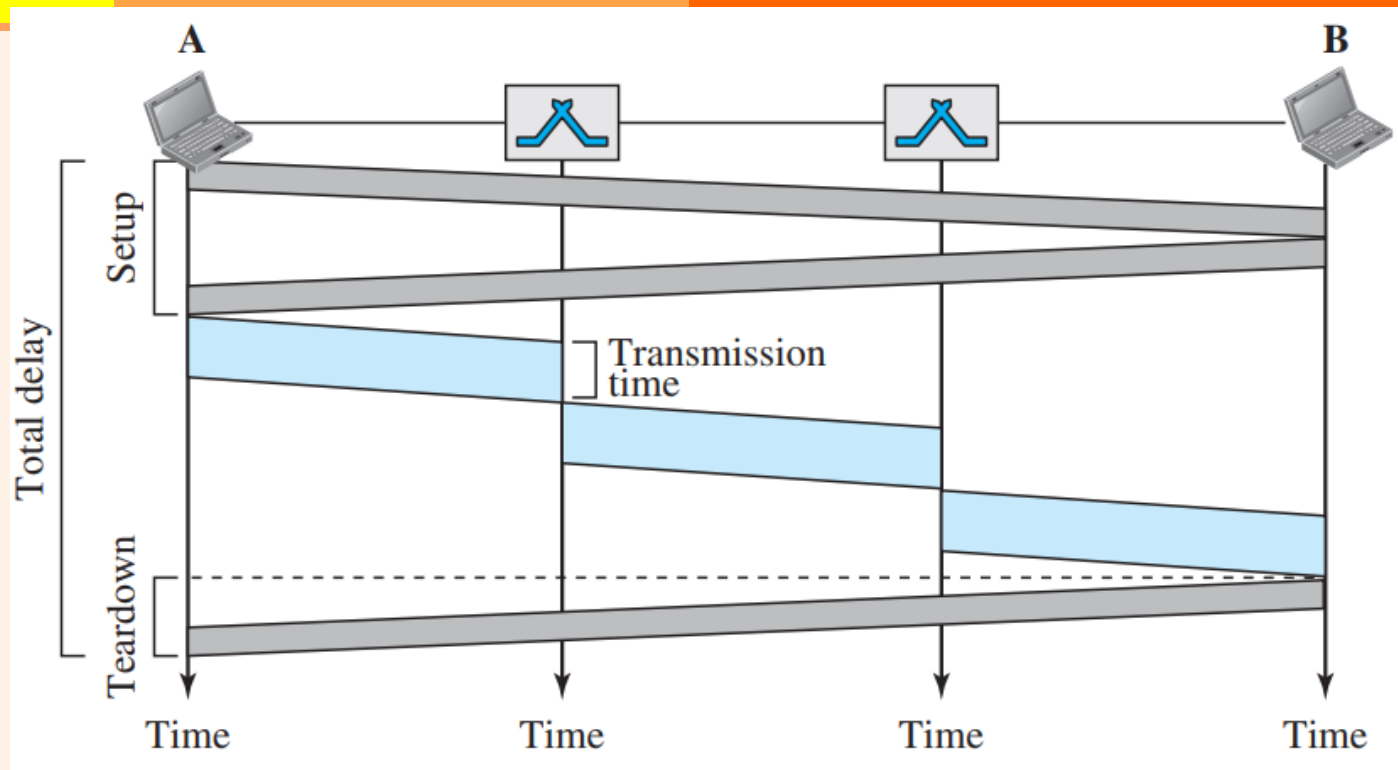


Figure: Source-to-destination data transfer in a virtual-circuit network

Delay in Virtual-Circuit Packet Switching



The packet is traveling through two switches (routers). There are three transmission times ($3T$), three propagation times (3τ), data transfer depicted by the sloping lines, a setup delay (which includes transmission and propagation in two directions), and a teardown delay (which includes transmission and propagation in one direction). We ignore the processing time in each switch. The total delay time is

$$\text{Total delay} = 3T + 3\tau + \text{setup delay} + \text{teardown delay}$$

Datagram vs Virtual-Circuits Switching

Datagram	Virtual circuits
It is connection less service.	Virtual circuits are connection-oriented.
More flexible: Routing can be used to avoid congested parts of the network	Packets are forwarded more quickly. No routing decisions to make
Datagram networks are not as reliable as Virtual Circuits.	Virtual Circuits are highly reliable.
Efficiency high, delay more	Efficiency low and delay less
It is always easy and cost-efficient to implement.	Implementation of virtual circuits is costly .
Requires more overhead per packet because each packet contains information about its destination address and other routing information.	Requires less overhead per packet because connection setup and state maintenance are done only once.



Books

1. Forouzan, B. A. "Data Communication and Networking. Tata McGraw." (2005).

References

1. Prakash C. Gupta, "Data communications", Prentice Hall India Pvt.
2. William Stallings, "Data and Computer Communications", Pearson
3. Forouzan, B. A. "Data Communication and Networking. Tata McGraw." (2005).